2025115098_assigment3

1) Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, i, largest;
    int *arr;

    printf("Enter n: ");
    scanf("%d", &n);

    arr = (int *)malloc(n * sizeof(int));

    printf("Enter %d elements: ", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    largest = *arr;

    for (i = 1; i < n; i++)
    {
        if (*(arr + i) > largest)
        {
            largest = *(arr + i);
        }
    }

    printf("\nLargest number: %d", largest);

    free(arr);

    return 0;
}
```

Output:

```
Enter n: 5
Enter 5 elements: 12 45 23 67 34

Largest number: 67
```

2) Print an array in reverse order using pointers.

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[10], n, i;
    int *p;

    printf("Enter n: ");
    scanf("%d", &n);

    p = arr;

    printf("Enter %d elements: ", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", p + i);
    }

    printf("\nArray in reverse order: ");

    for (i = n - 1; i >= 0; i--)
    {
        printf("%d ", *(p + i));
    }

    return 0;
}
```

Output:

```
Enter n: 5
Enter 5 elements: 10 20 30 40 50

Array in reverse order: 50 40 30 20 10
```

3) Write a C program to store information of students using structure. Sort items by name.

Source Code:

```
#include <stdio.h>
#include <string.h>

struct student
{
    int rollno;
    char name[20];
    float mark;
};

int main()
{
    struct student s[10], temp;
    int n, i, j;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("\n--- Student Details ---\n");

    for (i = 0; i < n; i++)
    {
        printf("\nRoll no: ");
        scanf("%d", &s[i].rollno);

        printf("Name: ");
        scanf("%s", s[i].name);

        printf("Mark: ");
        scanf("%f", &s[i].mark);
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (strcmp(s[i].name, s[j].name) > 0)
            {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    printf("\n--- Sorted by Name ---\n");

    for (i = 0; i < n; i++)
    {
        printf("\nRoll no: %d", s[i].rollno);
        printf("\nName: %s", s[i].name);
        printf("\nMark: %.2f\n", s[i].mark);
    }

    return 0;
}
```

Output:

```
Enter n: 3

--- Student Details ---

Roll no: 101
Name: Arun
Mark: 85

Roll no: 102
Name: Bala
Mark: 90

Roll no: 103
Name: Charan
```

```
Mark: 78

--- Sorted by Name ---

Roll no: 101
Name: Arun
Mark: 85.00

Roll no: 102
Name: Bala
Mark: 90.00

Roll no: 103
Name:  Charan
Mark: 78.00
```

4) Write a C program to add n distances given in inch-feet system using structures.

Source Code:

```
#include <stdio.h>

struct distance
{
    int feet;
    float inch;
};

int main()
{
    struct distance d[10], sum = {0, 0};
    int n, i;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("\n--- Enter Distances ---\n");

    for (i = 0; i < n; i++)
    {
        printf("\nDistance %d", i + 1);

        printf("\nFeet: ");
        scanf("%d", &d[i].feet);

        printf("Inch: ");
        scanf("%f", &d[i].inch);

        sum.feet += d[i].feet;
        sum.inch += d[i].inch;
    }

    if (sum.inch >= 12)
    {
        sum.feet += (int)(sum.inch / 12);
        sum.inch = sum.inch - ((int)(sum.inch / 12) * 12);
    }

    printf("\nTotal Distance: %d feet %.2f inches", sum.feet, sum.inch);

    return 0;
}
```

Output:

```
Enter n: 3

--- Enter Distances ---

Distance 1
Feet: 5
Inch: 8

Distance 2
Feet: 3
Inch: 6

Distance 3
Feet: 2
Inch: 10

Total Distance: 12 feet 0.00 inches
```

5) Write a C program to add two complex numbers by passing structures to a function.

Source Code:

```
#include <stdio.h>

struct complex
{
    float real;
    float imag;
};

struct complex add(struct complex c1, struct complex c2)
{
    struct complex result;

    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;

    return result;
}

int main()
{
    struct complex c1, c2, result;

    printf("Enter real and imaginary parts of first complex number: ");
    scanf("%f %f", &c1.real, &c1.imag);

    printf("Enter real and imaginary parts of second complex number: ");
    scanf("%f %f", &c2.real, &c2.imag);

    result = add(c1, c2);

    printf("\nSum = %.2f + %.2fi", result.real, result.imag);

    return 0;
}
```

Output:

```
Enter real and imaginary parts of first complex number: 4 5
Enter real and imaginary parts of second complex number: 3 2

Sum = 7.00 + 7.00i
```

6) Write a C program to maintain inventory management system using structures.

Source Code:

```
#include <stdio.h>

struct item
{
    int id;
    char name[20];
    int quantity;
    float price;
};

int main()
{
    struct item item[10];
    int n, i, choice, id, found;

    printf("Enter number of items: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nItem %d", i + 1);

        printf("\nItem ID: ");
        scanf("%d", &item[i].id);

        printf("Item Name: ");
        scanf("%s", item[i].name);

        printf("Quantity: ");
        scanf("%d", &item[i].quantity);

        printf("Price: ");
        scanf("%f", &item[i].price);
    }

    do
    {
        printf("\n--- Inventory Management ---\n");
        printf("1. Display Items\n");
        printf("2. Search Item\n");
        printf("3. Update Quantity\n");
        printf("4. Exit\n");

        printf("Enter choice: ");
        scanf("%d", &choice);

        if (choice == 1)
        {
            printf("\n--- Inventory Details ---\n");

            for (i = 0; i < n; i++)
            {
                printf("\nItem ID: %d", item[i].id);
                printf("\nName: %s", item[i].name);
                printf("\nQuantity: %d", item[i].quantity);
                printf("\nPrice: %.2f\n", item[i].price);
            }
        }
        else if (choice == 2)
        {
            printf("\nEnter Item ID: ");
            scanf("%d", &id);

            found = 0;

            for (i = 0; i < n; i++)
            {
                if (item[i].id == id)
                {
                    printf("\nItem Name: %s", item[i].name);
                    printf("\nQuantity: %d", item[i].quantity);
                    printf("\nPrice: %.2f\n", item[i].price);
                    found = 1;
                }
            }
```

```
            if (found == 0)
            {
                printf("\nItem not found.");
            }
        }
        else if (choice == 3)
        {
            printf("\nEnter Item ID: ");
            scanf("%d", &id);

            found = 0;

            for (i = 0; i < n; i++)
            {
                if (item[i].id == id)
                {
                    printf("Enter new quantity: ");
                    scanf("%d", &item[i].quantity);
                    printf("\nQuantity updated successfully.");
                    found = 1;
                }
            }

            if (found == 0)
            {
                printf("\nItem not found.");
            }
        }

    } while (choice != 4);

    return 0;
}
```

Output:

```
Enter number of items: 2

Item 1
Item ID: 101
Item Name: Pen
Quantity: 50
Price: 10

Item 2
Item ID: 102
Item Name: Book
Quantity: 20
Price: 50

--- Inventory Management ---
1. Display Items
2. Search Item
3. Update Quantity
4. Exit
Enter choice: 1

--- Inventory Details ---

Item ID: 101
Name: Pen
Quantity: 50
Price: 10.00

Item ID: 102
Name: Book
Quantity: 20
Price: 50.00

--- Inventory Management ---
1. Display Items
2. Search Item
3. Update Quantity
4. Exit
Enter choice: 4
```

7) Write a C program to convert a decimal number to binary, octal and hexadecimal.

Source Code:

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("\nBinary: ");

    if (n == 0)
    {
        printf("0");
    }
    else
    {
        int binary[32], i = 0, temp = n;

        while (temp > 0)
        {
            binary[i] = temp % 2;
            temp = temp / 2;
            i++;
        }

        while (i > 0)
        {
            i--;
            printf("%d", binary[i]);
        }
    }

    printf("\nOctal: %o", n);
    printf("\nHexadecimal: %X", n);

    return 0;
}
```

Output:

```
Enter a decimal number: 25

Binary: 11001
Octal: 31
Hexadecimal: 19
```

8) Write a C program to calculate difference between two time periods using structures.

Source Code:

```
#include <stdio.h>

struct time
{
    int hour;
    int minute;
    int second;
};

int main()
{
    struct time t1, t2, diff;

    printf("Enter first time (hour minute second): ");
    scanf("%d %d %d", &t1.hour, &t1.minute, &t1.second);

    printf("Enter second time (hour minute second): ");
    scanf("%d %d %d", &t2.hour, &t2.minute, &t2.second);

    if (t1.second < t2.second)
    {
        t1.second += 60;
        t1.minute--;
    }

    diff.second = t1.second - t2.second;

    if (t1.minute < t2.minute)
    {
        t1.minute += 60;
        t1.hour--;
    }

    diff.minute = t1.minute - t2.minute;
    diff.hour = t1.hour - t2.hour;

    printf("\nTime Difference: %d:%d:%d", diff.hour, diff.minute, diff.second);

    return 0;
}
```

Output:

```
Enter first time (hour minute second): 10 30 45
Enter second time (hour minute second): 8 20 30

Time Difference: 2:10:15
```

9) Write a C program to implement employee management system using structures.

Source Code:

```
#include <stdio.h>

struct employee
{
    int id;
    char name[20];
    float salary;
};

int main()
{
    struct employee e[10];
    int n, i, choice, id, found;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
    {
        printf("\nEmployee %d", i + 1);

        printf("\nEmployee ID: ");
        scanf("%d", &e[i].id);

        printf("Name: ");
        scanf("%s", e[i].name);

        printf("Salary: ");
        scanf("%f", &e[i].salary);
    }

    do
    {
        printf("\n--- Employee Management ---\n");
        printf("1. Display Employees\n");
        printf("2. Search Employee\n");
        printf("3. Exit\n");

        printf("Enter choice: ");
        scanf("%d", &choice);

        if (choice == 1)
        {
            printf("\n--- Employee Details ---\n");

            for (i = 0; i < n; i++)
            {
                printf("\nEmployee ID: %d", e[i].id);
                printf("\nName: %s", e[i].name);
                printf("\nSalary: %.2f\n", e[i].salary);
            }
        }
        else if (choice == 2)
        {
            printf("\nEnter Employee ID: ");
            scanf("%d", &id);

            found = 0;

            for (i = 0; i < n; i++)
            {
                if (e[i].id == id)
                {
                    printf("\nEmployee ID: %d", e[i].id);
                    printf("\nName: %s", e[i].name);
                    printf("\nSalary: %.2f\n", e[i].salary);
                    found = 1;
                }
            }

            if (found == 0)
            {
                printf("\nEmployee not found.");
            }
        }
```

```
    } while (choice != 3);

    return 0;
}
```

Output:

```
Enter number of employees: 2

Employee 1
Employee ID: 101
Name: Arun
Salary: 25000

Employee 2
Employee ID: 102
Name: Bala
Salary: 30000

--- Employee Management ---
1. Display Employees
2. Search Employee
3. Exit
Enter choice: 1

--- Employee Details ---

Employee ID: 101
Name: Arun
Salary: 25000.00

Employee ID: 102
Name: Bala
Salary: 30000.00

--- Employee Management ---
1. Display Employees
2. Search Employee
3. Exit
Enter choice: 3
```

10) Write a C program to calculate the area of rectangle and triangle using function pointers.

Source Code:

```
#include <stdio.h>

float rectangle(float length, float breadth)
{
    return length * breadth;
}

float triangle(float base, float height)
{
    return 0.5 * base * height;
}

int main()
{
    float length, breadth, base, height;
    float (*area)(float, float);

    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &length, &breadth);

    area = rectangle;

    printf("\nArea of rectangle: %.2f", area(length, breadth));

    printf("\n\nEnter base and height of triangle: ");
    scanf("%f %f", &base, &height);

    area = triangle;

    printf("\nArea of triangle: %.2f", area(base, height));

    return 0;
}
```

Output:

```
Enter length and breadth of rectangle: 10 5

Area of rectangle: 50.00

Enter base and height of triangle: 8 6

Area of triangle: 24.00
```